

INTERNSHIP REPORT

AR Anatomy and Engineering Visualization Learning Tool

Internship Organization:



Technical Career Education (TCE)

5 Floor, Sahyadri Campus, Adyar, Mangalore, Karnataka - 575007, India

Under the Guidance of:

Ms. Thejaswini

Department of Computer Science

Submitted by

Ananya Gopalakrishnan

Register No: **23BCGDR002**

BACHELOR OF COMPUTER APPLICATION

(Game Development and AR/VR) with TCS



YENEPOYA INSTITUTE OF ARTS, SCIENCE, COMMERCE & MANAGEMENT

(A constituent unit of Yenepoya Deemed to be University)

Deralakatte, Mangaluru

CERTIFICATE

This is to certify that the project work entitled “AR Anatomy and Engineering Visualization Learning Tool” has been successfully carried out at “Technical Career Education (TCE)” by Ananya Gopalakrishnan (Reg. No. 23BCGDR002), student of 3rd Year BCA (Game Development and AR/VR), under the supervision and guidance of Ms. Thejaswini, Department of Computer Science, Yenepoya Institute of Arts, Science, Commerce and Management. This dissertation is submitted in partial fulfilment for the award of the degree of Bachelor of Computer Application by Yenepoya (Deemed to be University) during the academic year 2025–26.

Internal Guide:

Chairperson

Ms. Thejaswini

Department of Computer Science

Internal Examiner:

External Examiner

PRINCIPAL

Dr. Jeevan Raj

**The Yenepoya Institute of Arts,
Science, Commerce and Management**
(Deemed to be University)

Submitted for the Viva-Voce

Place: Mangalore

Examination held on:

Date:

DECLARATION

I, **ANANYA GOPALAKRISHNAN**, a student of **BCA (specialization)** at **Yenepoya Institute of Arts, Science, Commerce, and Management, Balmatta, Mangalore**, affiliated with **Yenepoya (Deemed to be University)**, hereby declare that this internship report titled “ **AR anatomy and Engineering visualization Tool**” is a genuine and original record of the work undertaken by me as part of my academic curriculum.

This report documents the **knowledge, skills, and practical experience** acquired during my internship at Technical Career Education (TCE) It includes methodologies, analytical processes, and investigative approaches aligned with **recognized industry standards in digital forensics, incident response, and cybersecurity investigations**.

I extend my sincere gratitude to **[name], Internship Program Head**, for her/his **valuable guidance, mentorship, and support** throughout the internship period. I also express my appreciation to **(organization name)** for providing an enriching environment and exposure to real-world cybersecurity operations.

Furthermore, I acknowledge the support and encouragement received from my institutional guide, **Ms. Thejaswini** the faculty of **Yenepoya Institute of Arts, Science, Commerce, and Management**. Their insights and assistance have been instrumental in successfully completing this internship

Intern Signature:

Name: Ananya Gopalakrishnan

Date:

Internship Supervisor Signature:

Name: Pulkit Garg

Designation: Internship Program Head

Organization: TCE

Date: 30-04-2026

Institution Guide Signature:

Name: Ms. Thejaswini

Institute: Yenepoya Institute of Arts, Science, Commerce, and Management

Date:

Head of the Department (HOD) Signature:

Name: Dr. Rathnakar Shetty P

Head of Department of Computer Science

Institute: Yenepoya Institute of Arts, Science, Commerce, and Management

Date: 30-04-2026

ACKNOWLEDGEMENT

I sincerely express my gratitude to Yenepoya Institute of Arts, Science, Commerce, and Management, affiliated with Yenepoya (Deemed to be University), Mangalore, for providing me with the opportunity to undertake this internship as part of my academic curriculum.

I extend my heartfelt thanks to Dr. Jeevan Raj, Principal, for his continuous support and guidance in facilitating this learning opportunity. I also express my sincere appreciation to Dr. Rathnakara Shetty P, Head of Department (Computer Science), for his encouragement and academic support throughout my internship journey.

I am deeply grateful to Mr. Pulkit Garg, Mentor, Technical Career Education (TCE) at Sahyadri College of Engineering & Management, for his visionary leadership and for fostering a dynamic learning environment throughout the internship period. His guidance, expert feedback, and willingness to share practical industry knowledge were instrumental in helping me successfully execute this project during the internship.

I am also sincerely grateful to my institutional guide, Ms. Fathima Thoufeera, Department of Computer Science, Yenepoya Institute of Arts, Science, Commerce, and Management, for her academic mentorship and for providing valuable insights that helped bridge the gap between theoretical learning and practical application in the field of Augmented Reality and interactive application development..

I extend my appreciation to the team at Sahyadri College of Engineering & Management for providing access to the necessary hardware, tools, and workspace that enabled hands-on learning throughout the eight-week internship.

Lastly, I am grateful to my faculty mentors, family, and peers for their encouragement and motivation throughout this journey. This internship has been a transformative experience and has significantly deepened my understanding of Augmented Reality, real-time interaction systems, and collaborative application development.

Name: Ananya Gopalakrishnan

Reg NO: 23BCGDR002

Date: 30-04-2026

INTERNSHIP OFFER LETTER



Date: 13-02-2026

INTERNSHIP OFFER LETTER

To:

**Ananya Gopalakrishnan ,
Yenepoya College**

Dear **Ananya Gopalakrishnan ,**

We are pleased to offer you an internship opportunity with **Technical Career Education TCE**. The internship is scheduled for a period of **2 Months**, commencing from **02-03-2026 to 02-05-2026**. You are required to report in person on **02-03-2026**. Your designated mentor for the duration of the internship will be **Jason D'Silva - Program Mentor**

Below are the terms and conditions of your engagement.

Internship Details

- **Internship Allowance: Unpaid Internship**
- **Tenure: 2 Months**
- **Start and End Dates: 02-03-2026 to 02-05-2026**
- **Project Title: VR Digital Twin – Mphasis Lab & InUnity Office Reconstruction**
- **Deliverables:**

Photogrammetry scan of Mphasis Lab and InUnity Office (cleaned mesh); Optimized 3D environment in Unity (<500K triangles in view); VR teleportation + continuous locomotion system; Interactive objects (clickable equipment, openable doors, info panels); World-space UI canvases showing lab data, team info, project descriptions; Deployed Meta Quest APK running at 72 FPS; User testing report (3–5 participants); Demo video; GitHub repository; Technical report/PPT; Stretch: WebXR browser version via A-Frame

Terms of Engagement

Please note that this internship is a training opportunity and does not constitute an offer of employment. Completion of the internship does not guarantee a job with the company. However, interns who demonstrate strong technical and professional capabilities may be considered for future roles, including TCE roles, based on performance evaluations.

You are expected to proactively learn new technologies and complete assigned tasks and projects to the best of your ability. Upon successful completion of the internship and assigned deliverables, you will be awarded a Project Completion Certificate.



Date: 13-02-2026

During your internship, you may be granted access to proprietary and confidential information. By accepting this offer, you agree to maintain the confidentiality of such information and not to use or disclose it outside the scope of your internship. At the end of the internship, you must return all company property, equipment, and documentation, including digital files and materials.

Additional Guidelines

- This internship is offered on an at-will basis, meaning that either party may terminate the internship at any time and for any reason.
- The terms outlined in this letter are subject to change at the discretion of the Company.
- The internship may be extended up to six months, depending on your performance.
- You are required to develop a project proposal in collaboration with your mentor, as per the attached Addendum.
- You must follow the mentor's instructions and actively manage the project.
- The internship will involve a hybrid format (both virtual and in-person). Travel arrangements for in-person sessions are your responsibility.
- A draft project report and presentation must be submitted to an assigned review panel. Revisions should be made based on feedback received.
- A final approved project submission is a prerequisite for receiving the Internship Certificate.

By accepting this offer, you agree to comply with all company policies applicable to non-employee interns, including but not limited to the anti-harassment policy.

This letter constitutes the full agreement between you and TCE regarding the internship. Any changes must be documented in writing and signed by both parties.

To confirm your acceptance, please sign and return a copy of this letter to the HR department.



Date: 13-02-2026

We look forward to your contributions and hope that your internship with us will be a valuable and rewarding experience.

Best regards,
TCE



A handwritten signature in blue ink, appearing to read "Johnson".

Mr. Johnson Tellis

Employee Acceptance Signature

CEO, TCE – Technical Career Education



INTERNSHIP COMPLETION CERTIFICATE

COMPANY PROFILE: Technical Career Education (TCE)

Technical Career Education (TCE), Bangalore, is a leading organization dedicated to providing industry-oriented training in emerging technologies such as Augmented Reality (AR), Virtual Reality (VR), Robotics, Artificial Intelligence (AI), and Digital Twin systems. The organization focuses on bridging the gap between academic learning and industry requirements by offering practical exposure through structured internship programs and real-time project development.

TCE is committed to enhancing the technical capabilities of students by providing hands-on training using modern tools and technologies. The organization offers specialized training in Unity development, AR Foundation, 3D modeling, and application development, enabling students to gain in-depth knowledge and practical skills required in the field of immersive technologies.

The internship program at TCE is designed to provide a comprehensive learning experience that includes technical sessions, guided project work, and performance evaluation. Students are encouraged to work on real-world applications, collaborate with peers, and apply theoretical knowledge to practical scenarios. This approach helps in developing problem-solving abilities, creativity, and technical proficiency.

TCE also emphasizes innovation and continuous learning by exposing students to the latest trends and advancements in technology. The organization provides a supportive and collaborative environment where students can experiment, learn, and grow professionally. Regular mentoring sessions and project reviews ensure that students receive proper guidance throughout their internship period.

During the internship, I had the opportunity to work on an Augmented Reality-based educational application, which helped me understand the practical implementation of AR technologies. The experience gained at Technical Career Education (TCE) has significantly contributed to my technical development and prepared me for future career opportunities in the field of AR/VR and software development.



ROLES AND RESPONSIBILITIES

1. Project: AR Anatomy and Engineering Visualization Learning Tool

Developed an Augmented Reality (AR) based educational application using Unity and AR Foundation to visualize complex anatomical and mechanical structures in a real-world environment.

1.1 AR Application Development

- Designed and developed the application using Unity 3D platform.
- Configured AR Foundation for plane detection, object placement, and tracking.
- Implemented AR session management and optimized real-time rendering performance.

1.2 3D Model Integration

- Imported and integrated 3D models such as human heart, kidney, and jet engine.
- Adjusted scaling, positioning, and orientation of models for accurate AR visualization.
- Ensured smooth rendering and optimized models for mobile performance.

1.3 User Interaction and Controls

- Implemented gesture-based interactions including tap to place, drag to rotate, and pinch to scale.
- Developed interactive buttons for model selection (Heart, Kidney, Engine).
- Added reset functionality to reposition models and restart the AR session.

1.4 User Interface (UI) Design

- Designed a simple and user-friendly UI using Unity Canvas system.
- Created buttons and text panels for displaying model descriptions.
- Fixed UI overlap issues and ensured proper alignment across different screen sizes.

1.5 Testing and Debugging

- Tested the application in real-time using Android device.
- Identified and resolved issues related to AR tracking and model placement instability.
- Improved performance to ensure smooth interaction and responsiveness.

1.6 Optimization and Performance Improvement

- Optimized application for better FPS and reduced lag in AR environment.
- Managed memory usage and ensured efficient handling of assets.
- Enhanced application stability during continuous usage.

2.Additional Activities and Learning

2.1 Git and GitHub Learning

- Attended sessions on Git version control system.
- Learned basic commands such as commit, push, pull, and branch management.
- Understood the importance of version control in collaborative development.

2.2 Industrial Exposure (TATA Visit)

- Participated in an industrial visit to observe real-world project implementation.
- Gained knowledge about industry workflow and professional development practices.
- Understood the importance of teamwork, communication, and project management.

3.Communication and Documentation

3.1 Documentation

- Maintained structured documentation of project development stages.
- Recorded implementation steps, challenges, and solutions.

3.2 Team Interaction

- Participated in discussions and mentoring sessions.
- Improved communication and teamwork skills through collaboration.

4.Overall Contribution

- Successfully developed a functional AR application.
- Gained hands-on experience in AR/VR development.
- Improved technical, problem-solving, and analytical skills.
- Completed the project within the given timeline with proper guidance.

LEARNING OUTCOMES

1. Augmented Reality (AR) Application Development

- Gained hands-on experience in developing AR applications using Unity and AR Foundation, understanding core concepts such as plane detection, object placement, and real-time environment tracking.
- Learned how AR systems integrate virtual objects into real-world environments, improving visualization and user interaction capabilities.

2. Unity Development and 3D Environment Handling

- Developed proficiency in Unity 3D for building interactive applications, including scene management, asset integration, and scripting.
- Understood the importance of optimization techniques such as reducing polygon count and efficient asset handling to ensure smooth performance on mobile devices.

3. 3D Model Integration and Visualization

- Gained experience in importing, scaling, and positioning 3D models such as the human heart, kidney, and jet engine within an AR environment.
- Learned how to enhance visual clarity and accuracy of models to improve educational understanding and user experience.

4. User Interaction and Gesture Control

- Implemented gesture-based interactions such as tap, drag, and pinch, improving the usability of the application.
- Understood how user interaction design plays a critical role in creating intuitive and engaging AR applications.

5. User Interface (UI) Design and Implementation

- Designed and developed user-friendly interfaces using Unity Canvas, including buttons, text panels, and navigation controls.
- Learned how to resolve UI alignment and overlapping issues across different mobile screen sizes.

6. Testing, Debugging, and Performance Optimization

- Gained practical experience in testing AR applications on real devices and identifying issues related to tracking, placement, and performance.
- Improved debugging skills by resolving errors and optimizing the application for better stability and responsiveness.

7.Version Control using Git and GitHub

- Learned the fundamentals of Git version control system, including repository management, commits, branching, and collaboration.
- Understood the importance of maintaining version history and managing code efficiently in team-based projects.

8.Industry Exposure and Practical Knowledge

- Gained real-world exposure through industrial visit (TATA), understanding how projects are developed and managed in professional environments.
- Learned about industry standards, teamwork, and workflow practices followed in software development.

9.Problem-Solving and Critical Thinking

- Developed analytical thinking and problem-solving skills while handling real-time project challenges.
- Learned how to identify issues, analyze root causes, and implement effective solutions systematically.

10.Communication and Professional Development

- Improved communication skills through regular discussions, mentoring sessions, and project presentations.
- Gained confidence in explaining technical concepts and working collaboratively in a professional environment.

11.Overall Learning Experience

- Successfully applied theoretical knowledge to practical implementation in AR/VR development.
- Enhanced technical, creative, and professional skills required for future career opportunities in immersive technologies and software development.

WEEKLY INTERNSHIP LOG

INTERN WEEKLY REPORT

Intern: Ananya Gopalakrishnan

Week Beginning: 02/03/2026 To: 06/03/2026

Position: Intern (AR/VR Development)

Employer: Technical Career Education (TCE)

Internship Supervisor's Name: Mr. Pulkit Garg

Internship Coordinator: Ms. Suzana

DAY	TIME IN	TIME OUT	TOTAL HOURS	DUTIES PERFORMED
MONDAY	9.30	4.30	6	Introduction to project, requirement analysis.
TUESDAY	9.30	4.30	6	Installed Unity, Android Studio setup
WEDNESDAY	9.30	4.30	6	Learned Unity basics
THURSDAY	9.30	4.30	6	AR Foundation setup
FRIDAY	9.30	4.30	6	Plane detection testing
SATURDAY	0	0	0	Off
SUNDAY	0	0	0	Off
TOTAL HOURS			30	

INTERN WEEKLY REPORT

Intern: Ananya Gopalakrishnan

Week Beginning: 09/03/2026 To: 13/03/2026

Position: Intern (AR/VR Development)

Employer: Technical Career Education (TCE)

Internship Supervisor's Name: Mr. Pulkit Garg

Internship Coordinator: Ms. Suzana

DAY	TIME IN	TIME OUT	TOTAL HOURS	DUTIES PERFORMED
MONDAY	9.30	4.30	6	Import 3D models
TUESDAY	9.30	4.30	6	AR model placement
WEDNESDAY	9.30	4.30	6	Scaling & Rotation
THURSDAY	9.30	4.30	6	Gesture basics
FRIDAY	9.30	4.30	6	Testing
SATURDAY	0	0	0	Off
SUNDAY	0	0	0	Off
TOTAL HOURS			30	

INTERN WEEKLY REPORT

Intern: Ananya Gopalakrishnan

Week Beginning: 16/03/2026 To: 20/03/2026

Position: Intern (AR/VR Development)

Employer: Technical Career Education (TCE)

Internship Supervisor's Name: Mr. Pulkit Garg

Internship Coordinator: Ms. Suzana

DAY	<u>TIME</u> <u>IN</u>	<u>TIME</u> <u>OUT</u>	<u>TOTAL</u> <u>HOURS</u>	<u>DUTIES PERFORMED</u>
MONDAY	9.30	4.30	6	UI Design
TUESDAY	9.30	4.30	6	Buttons & panels
WEDNESDAY	9.30	4.30	6	Model selection
THURSDAY	9.30	4.30	6	Annotation system
FRIDAY	9.30	4.30	6	Debugging UI
SATURDAY	0	0	0	Off
SUNDAY	0	0	0	Off
TOTAL HOURS			30	

INTERN WEEKLY REPORT

Intern: Ananya Gopalakrishnan

Week Beginning: 23/03/2026 To: 27/03/2026

Position: Intern (AR/VR Development)

Employer: Technical Career Education (TCE)

Internship Supervisor's Name: Mr. Pulkit Garg

Internship Coordinator: Ms. Suzana

DAY	TIME IN	TIME OUT	TOTAL HOURS	DUTIES PERFORMED
MONDAY	9.30	4.30	6	Add organ models
TUESDAY	9.30	4.30	6	Add engine models
WEDNESDAY	9.30	4.30	6	Optimization
THURSDAY	9.30	4.30	6	Performance testing
FRIDAY	9.30	4.30	6	Bug fixing
SATURDAY	0	0	0	Off
SUNDAY	0	0	0	Off
TOTAL HOURS			30	

INTERN WEEKLY REPORT

Intern: Ananya Gopalakrishnan

Week Beginning: 30/03/2026 To: 3/04/2026

Position: Intern (AR/VR Development)

Employer: Technical Career Education (TCE)

Internship Supervisor's Name: Mr. Pulkit Garg

Internship Coordinator: Ms. Suzana

DAY	TIME IN	TIME OUT	TOTAL HOURS	DUTIES PERFORMED
MONDAY	9.30	4.30	6	Gesture Improvements
TUESDAY	9.30	4.30	6	Reset button
WEDNESDAY	9.30	4.30	6	Anchor stability fix
THURSDAY	9.30	4.30	6	Model placement fix
FRIDAY	9.30	4.30	6	Testing
SATURDAY	0	0	0	Off
SUNDAY	0	0	0	Off
TOTAL HOURS			30	

INTERN WEEKLY REPORT

Intern: Ananya Gopalakrishnan

Week Beginning: 06/04/2026 To: 10/04/2026

Position: Intern (AR/VR Development)

Employer: Technical Career Education (TCE)

Internship Supervisor's Name: Mr. Pulkit Garg

Internship Coordinator: Ms. Suzana

DAY	TIME IN	TIME OUT	TOTAL HOURS	DUTIES PERFORMED
MONDAY	9.30	4.30	6	UI improvements
TUESDAY	9.30	4.30	6	Module integration
WEDNESDAY	9.30	4.30	6	Full testing
THURSDAY	9.30	4.30	6	Bug fixing
FRIDAY	9.30	4.30	6	Demo preparation
SATURDAY	0	0	0	Off
SUNDAY	0	0	0	Off
TOTAL HOURS			30	

INTERN WEEKLY REPORT

Intern: Ananya Gopalakrishnan

Week Beginning: 13/04/2026 To: 17/04/2026

Position: Intern (AR/VR Development)

Employer: Technical Career Education (TCE)

Internship Supervisor's Name: Mr. Pulkit Garg

Internship Coordinator: Ms. Suzana

DAY	TIME IN	TIME OUT	TOTAL HOURS	DUTIES PERFORMED
MONDAY	0	0	0	Leave
TUESDAY	0	0	0	Leave
WEDNESDAY	0	0	0	Leave
THURSDAY	0	0	0	Leave
FRIDAY	0	0	0	Leave
SATURDAY	0	0	0	Off
SUNDAY	0	0	0	Off
TOTAL HOURS			0	

INTERN WEEKLY REPORT

Intern: Ananya Gopalakrishnan

Week Beginning: 20/04/2026 To: 24/04/2026

Position: Intern (AR/VR Development)

Employer: Technical Career Education (TCE)

Internship Supervisor's Name: Mr. Pulkit Garg

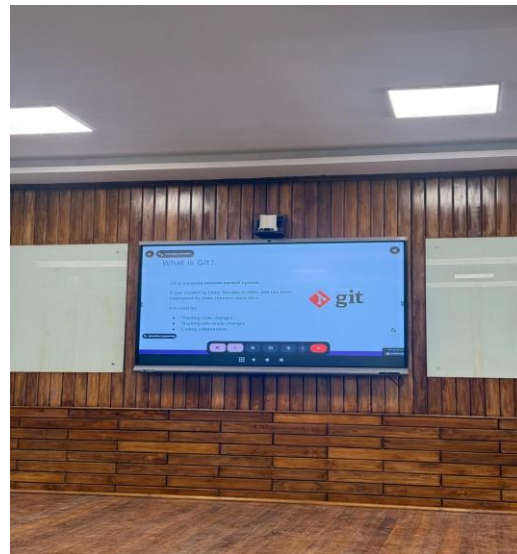
Internship Coordinator: Ms. Suzana

DAY	TIME IN	TIME OUT	TOTAL HOURS	DUTIES PERFORMED
MONDAY	9.30	4.30	6	Work review
TUESDAY	9.30	4.30	6	Final testing
WEDNESDAY	9.30	4.30	6	Documentation
THURSDAY	9.30	4.30	6	Report preparation
FRIDAY	9.30	4.30	6	Final corrections
SATURDAY	0	0	0	Off
SUNDAY	0	0	0	Off
TOTAL HOURS			30	

CERTIFICATION & OTHER ACTIVITIES

1.Git and GitHub Session

During the internship, I attended a session on Git and GitHub where I learned the fundamentals of version control systems. The session covered important concepts such as repository creation, commit, push, pull, and branch management. This helped me understand how developers collaborate and manage code efficiently in real-world projects.



2.Industrial Interaction – TATA Company Visit to Lab

During the internship period, professionals from TATA visited our lab to observe and interact with students and their ongoing projects. This visit provided a valuable opportunity to present our work in front of industry experts and gain practical insights.

I demonstrated my project titled “AR Anatomy and Engineering Visualization Learning Tool”, explaining its features such as AR-based model placement, interaction controls, and real-time visualization of 3D models like the heart, kidney, and jet engine.

The visiting team observed the working of the application and provided useful feedback on improving performance, user interface, and stability. They also shared insights on industry expectations and real-world applications of AR/VR technologies.

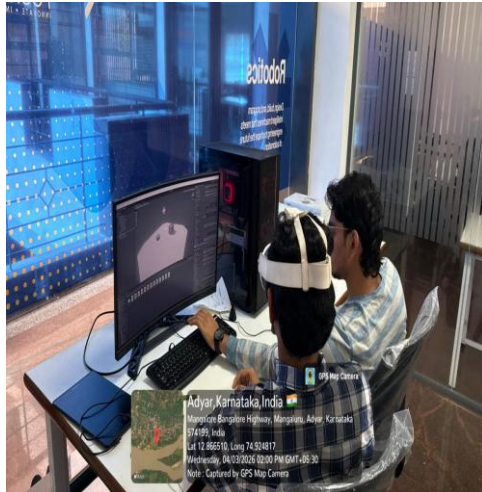
The interaction also helped me understand the importance of presenting projects clearly and effectively. I learned how to explain technical concepts in a simple and structured manner, which improved my communication and presentation skills.

In addition, the visit provided exposure to real-world industry practices and standards. It motivated me to enhance my project further and align my work with industry requirements and expectations.



3. Classroom and Technical Activities

I also participated in classroom-based technical sessions, including presentations and discussions on development tools such as Git and Unity. These sessions enhanced my theoretical understanding and practical application skills



4. Practical Learning Experience

Apart from project development, I gained hands-on experience in Unity, AR Foundation, and mobile-based AR applications. I also improved my debugging, UI design, and performance optimization skills.

Overall, these activities contributed significantly to my technical growth, communication skills, and professional readiness.

ABSTRACT

This project presents an Augmented Reality (AR)-based application designed to enhance learning by visualizing complex anatomical and engineering structures. The application enables users to interact with 3D models such as the human heart, kidney, and jet engine in a real-world environment using a mobile device.

The system is developed using Unity 3D and AR Foundation, which provide the necessary tools for implementing AR features such as plane detection, object placement, and real-time interaction. The application allows users to place, rotate, and scale 3D models, making the learning process more interactive, engaging, and easier to understand.

The main objective of this project is to improve the understanding of complex structures through visualization and interaction. Traditional learning methods often rely on static images and theoretical explanations, which may not effectively convey detailed structures. This application addresses that limitation by providing an immersive and practical learning experience.

The project also focuses on developing a user-friendly interface that allows easy navigation and interaction. Special attention is given to performance optimization to ensure smooth operation on mobile devices, including efficient handling of 3D models and stable AR tracking.

The application was tested under different conditions to ensure accuracy, responsiveness, and reliability. Feedback obtained during testing and industry interaction sessions helped in refining the system and improving overall performance.

Overall, this project demonstrates the potential of Augmented Reality in education by providing an effective and interactive platform for learning. It highlights how emerging technologies can be integrated into academic environments to enhance understanding and engagement.

The system can be further extended by adding more models, incorporating audio-visual explanations, and supporting advanced technologies such as AR glasses and multi-user collaboration for an improved learning experience.

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1. INTRODUCTION

1.1 Background

Augmented Reality (AR) and Virtual Reality (VR) are rapidly evolving technologies that are transforming the way users interact with digital systems. Unlike traditional applications that rely on static screens, AR integrates virtual elements into the real-world environment, enabling interactive and immersive experiences. This has made AR a key technology in fields such as education, healthcare, gaming, and industrial training.

In the domain of game development, AR enhances user engagement by combining real-world surroundings with virtual content. Users are no longer passive viewers; instead, they actively interact with digital objects placed in their environment. This creates a more intuitive and engaging experience compared to conventional interfaces.

The development of AR applications has become more accessible with tools like Unity, AR Foundation, and ARCore. These platforms provide essential capabilities such as plane detection, object tracking, and environmental understanding. As a result, developers can build applications that respond intelligently to real-world conditions.

This internship project focuses on exploring these technologies through practical implementation. It aims to build an AR-based system that demonstrates real-time interaction, efficient performance, and user-friendly design, while also providing exposure to industry-level tools and workflows.

1.2 Objectives

The main objective of this internship is to gain practical knowledge and hands-on experience in Game Development and AR/VR technologies. The project emphasizes both technical implementation and understanding of real-world applications.

Key objectives include:

- Developing an interactive AR application using Unity and AR Foundation
- Understanding ARCore features such as motion tracking and surface detection
- Implementing object placement, scaling, and interaction in real time
- Learning C# scripting for controlling application behavior

- Using Git and GitHub for version control and project management
- Enhancing problem-solving and debugging skills
- Improving teamwork, communication, and presentation abilities

Additionally, the project aims to bridge the gap between theoretical knowledge and practical application by working on real-time scenarios and industry-relevant tasks.

1.3 Scope

The project focuses on designing and developing an Augmented Reality application that allows users to visualize and interact with 3D objects in their real-world environment. The system is developed using Unity and AR Foundation, ensuring compatibility with Android devices that support ARCore.

The application includes several core functionalities:

- Detection of flat surfaces using ARCore
- Placement of virtual objects in the real world
- Interaction features such as rotation, scaling, and repositioning
- Basic user interface for controlling actions
- Integration of tracking and anchoring mechanisms

The project is limited to a prototype-level implementation and does not include advanced features such as cloud-based synchronization, high-end graphics optimization, or cross-platform VR integration. However, it provides a strong foundation for future development and scalability.

1.4 Problem Definition

Traditional applications often fail to provide an interactive and immersive user experience. Static content such as text and images does not effectively convey complex concepts, especially in areas that require visualization and real-time interaction.

Although AR technology offers a solution, existing systems face several challenges:

- Limited tracking accuracy in certain environments

- High latency and performance issues
- Lack of intuitive user interfaces
- Dependency on high-end devices
- Stability issues during continuous usage

These limitations reduce accessibility and user experience. This project aims to overcome these issues by developing a lightweight, stable, and user-friendly AR application that ensures smooth interaction and efficient performance.

2. TOOLS AND TECHNOLOGY USED

2.1 Unity Game Engine

Unity is a powerful and widely used game development platform that enables the creation of interactive 2D, 3D, AR, and VR applications. It provides a complete development environment with tools for rendering, animation, scripting, and user interface design.



Key features of Unity include:

- Cross-platform development support
- Built-in physics and rendering engine
- Large asset store for resources
- Support for C# scripting
- Easy integration with AR technologies

Unity simplifies the development process and allows developers to focus on building interactive experiences.

2.2 AR Foundation

AR Foundation is a framework that enables developers to build AR applications using a unified workflow. It connects Unity with platform-specific AR systems such as ARCore.

It provides essential features such as:

- Plane detection for identifying surfaces
- Raycasting for accurate object placement
- Anchor support for stable positioning
- Environment tracking

This framework ensures compatibility across different devices and platforms.

2.3 ARCore

ARCore is a platform developed by Google for creating AR applications on Android devices. It allows devices to understand their surroundings and interact with virtual objects.

Main capabilities include:

- Motion tracking
- Environmental understanding
- Light estimation

These features help create realistic and immersive AR experiences.

2.4 Programming Language (C#)

C# is used in Unity for scripting and implementing application logic. It is an object-oriented programming language that helps in creating structured and efficient code.

In this project, C# is used for:

- Managing user interactions
- Controlling object behavior
- Implementing AR functionalities
- Handling UI elements

2.5 Version Control (Git & GitHub)

Version control is essential for managing project development. Git is used to track changes in code, while GitHub provides a platform for collaboration and storage.

Advantages include:

- Tracking project history
- Managing code versions
- Supporting teamwork
- Ensuring data backup

2.6 XR Devices (XREAL Glasses)

XREAL glasses are advanced AR wearable devices that provide an immersive augmented reality experience. These glasses project virtual content into the user's field of view, allowing interaction with digital objects in real time.

In this project, XREAL glasses represent the future scope of AR visualization, where users can experience AR content without relying on mobile screens. These devices enhance user experience by providing hands-free interaction and improved immersion.

The integration of AR applications with devices like XREAL glasses can significantly improve usability in fields such as education, healthcare, and industrial training.



3. DATA COLLECTION AND ANALYSIS

3.1 Data Collection

Data collection is a fundamental step in the development of an Augmented Reality (AR) application. It involves gathering all necessary resources required for building and testing the system. In this project, data mainly consists of 3D models, textures, UI elements, and reference materials.

The 3D models used in the application, such as anatomical parts and engineering components, were collected from reliable sources including the Unity Asset Store and open-source libraries. While selecting these assets, attention was given to factors such as model quality, file size, and compatibility with mobile devices.

In addition to visual assets, learning resources such as documentation, tutorials, and technical guides were also collected. These materials helped in understanding AR concepts, Unity development, and implementation techniques.

Key aspects considered during data collection:

- Relevance of models to the project objectives
- Optimization level of assets for mobile performance
- Compatibility with Unity and AR Foundation
- Availability of supporting textures and materials

Proper data collection ensures that the development process is smooth and that the final application performs efficiently.

3.2 Data Preprocessing

After collecting the required data, preprocessing is carried out to prepare the assets for integration into the AR environment. This step is essential for improving performance and ensuring smooth rendering on mobile devices.

Preprocessing involves optimizing 3D models and adjusting their properties to match system requirements. Since AR applications run in real time, unoptimized models can cause lag and reduce user experience.

Main preprocessing steps include:

- Reducing polygon count to improve rendering speed

- Adjusting scale and orientation of models
- Compressing textures to reduce memory usage
- Removing unnecessary components and hidden elements
- Testing models inside Unity for compatibility

These optimizations help in maintaining a balance between visual quality and performance. Proper preprocessing also reduces loading time and ensures stable execution of the application.

3.3 System Components and Data Flow

The AR application consists of multiple components that work together to provide a seamless interactive experience. Each component plays a specific role in processing and displaying data.

The main components of the system include:

- **Camera Input:** Captures the real-world environment
- **ARCore Processing:** Detects surfaces and tracks movement
- **Unity Engine:** Renders virtual objects
- **User Interface:** Allows interaction with the system

The data flow within the system follows a continuous cycle:

1. The device camera captures real-world surroundings
2. ARCore analyzes the environment and detects surfaces
3. Unity processes the data and renders virtual objects
4. The user interacts with objects through touch gestures
5. The system updates object position and maintains stability

This structured flow ensures real-time interaction and smooth communication between hardware and software components.

3.4 Analysis of AR Environment

The effectiveness of an AR application depends on how well it adapts to the surrounding environment. The system continuously analyzes environmental conditions to ensure accurate object placement and stable interaction.

Several factors influence AR performance:

- Lighting conditions
- Surface texture and flatness
- Device movement and orientation
- Camera quality and sensor accuracy

A well-lit environment improves tracking accuracy, while textured surfaces help in better plane detection. Sudden movements or poor lighting may reduce stability and affect user experience.

The system is designed to handle these conditions by adjusting object placement and maintaining tracking consistency. This ensures that virtual objects remain stable and appear realistic within the real-world environment.

Overall, proper analysis of the AR environment plays a crucial role in delivering a smooth, reliable, and immersive user experience.

4. SYSTEM REQUIREMENTS AND ANALYSIS

4.1 Functional Requirements

Functional requirements define the core features and operations that the AR application must perform. These requirements ensure that the system meets user needs and provides the intended functionality.

The application is designed to allow users to interact with virtual objects in a real-world environment. It must accurately detect surfaces and enable smooth object placement and manipulation.

Key functional requirements include:

- Detect real-world surfaces using ARCore
- Place virtual 3D objects on detected surfaces
- Allow user interaction such as rotation, scaling, and movement
- Display relevant information about selected objects
- Provide options to reset or remove objects
- Support continuous tracking and real-time updates

These functionalities form the foundation of the system and ensure that the application operates effectively. Proper implementation of these features improves usability and enhances user experience.

4.2 Non-Functional Requirements

Non-functional requirements describe how the system performs under different conditions. These requirements focus on performance, reliability, and user experience rather than specific features.

Important non-functional requirements include:

- **Performance:** The application should run smoothly with minimal lag and fast response time
- **Usability:** The interface should be simple and easy to understand
- **Reliability:** The system should function consistently without crashes

- **Scalability:** The application should allow future enhancements
- **Compatibility:** It should work on supported Android devices

These requirements ensure that the application provides a stable and efficient experience for users. Meeting these conditions is essential for maintaining quality and user satisfaction.

4.3 Hardware Requirements

The AR application requires specific hardware components to function effectively. Since AR relies heavily on camera input and sensor data, the device must meet certain minimum specifications.

Recommended hardware requirements:

- Android smartphone with ARCore support
- Minimum 4GB RAM for smooth performance
- Camera for capturing real-world environment
- Motion sensors (accelerometer and gyroscope)
- Sufficient storage for application and assets

Using appropriate hardware ensures better tracking accuracy and improves the overall performance of the application.

4.4 Software Requirements

The development and execution of the application depend on several software tools and frameworks. These tools provide the necessary environment for building, testing, and running the AR application.

Software requirements include:

- **Unity Game Engine:** Used for designing and developing the application
- **AR Foundation:** Provides AR functionalities such as plane detection and tracking
- **ARCore SDK:** Enables environment understanding and motion tracking
- **Visual Studio:** Used for writing and debugging C# code

- **Git and GitHub:** Used for version control and project management

These tools work together to support the complete development lifecycle of the application.

4.5 System Overview

The system integrates real-world input with virtual elements to create an interactive AR environment. It uses the device camera to capture surroundings and processes the data to detect flat surfaces.

Once a surface is detected, virtual objects are placed on it and rendered using Unity. The user can interact with these objects through touch gestures such as tapping, dragging, and pinching.

The system is designed to ensure:

- Real-time response to user actions
- Smooth interaction with minimal delay
- Stable object placement during movement
- Easy navigation and control

This overall structure makes the application efficient, user-friendly, and suitable for practical use. The system architecture also allows future enhancements and scalability.

5. IMPLEMENTATION

5.1 System Design

The system is designed to provide a seamless interaction between real-world environments and virtual objects using Augmented Reality (AR). It follows a structured architecture consisting of input, processing, and output layers.

The **input layer** captures real-world data through the device camera. The **processing layer**, powered by ARCore and Unity, analyzes the captured data to detect surfaces and track device movement. The **output layer** renders 3D objects and displays them in the real-world environment.

The design focuses on simplicity, efficiency, and usability. It ensures that users can easily interact with the system without requiring technical knowledge. The modular structure of the system also allows easy maintenance and future enhancements.

Key design considerations include:

- Real-time performance and responsiveness
- Stable object placement
- User-friendly interface
- Efficient resource utilization

5.2 Workflow / Process

The working of the AR application follows a clear and logical sequence of steps. This workflow ensures smooth interaction and easy understanding for users.

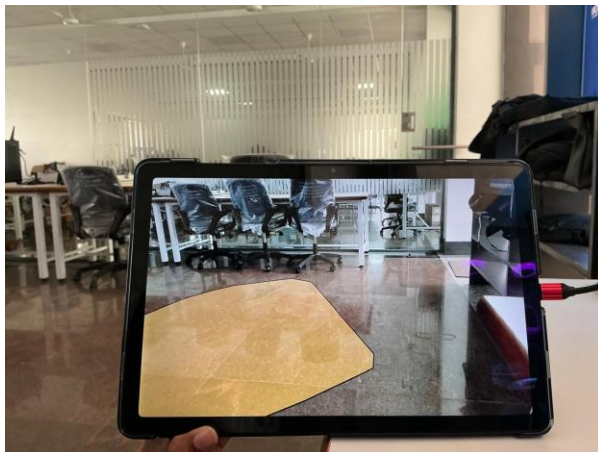
The process includes:

1. The user launches the application
2. The AR session is initialized

3. The device camera scans the environment



4. The system detects flat surfaces



5. The user selects and places a virtual object



6. The user interacts with the object using gestures



7. The system continuously updates object position

This step-by-step workflow ensures that the application operates efficiently and provides a smooth user experience. Each stage is designed to minimize delays and maintain stability.

5.3 Module Description

5.3.1 AR Model Interaction Module

This module is responsible for enabling interaction between the user and virtual objects. It allows users to manipulate objects using touch gestures such as tap, drag, and pinch.

Features of this module include:

- Rotation of objects in different directions
- Scaling objects based on user input
- Dragging objects across surfaces
- Real-time response to gestures

This module plays a key role in improving user engagement and interactivity.

5.3.2 QR Code Module

The QR Code module enables users to scan QR codes to access specific AR content. This simplifies navigation and allows quick loading of predefined models or scenes.

Benefits of this module include:

- Faster access to content
- Easy linking between physical and digital systems
- Reduced manual selection

This feature enhances usability and provides flexibility in accessing AR content.

5.3.3 AR Tracking Module

The AR Tracking module ensures accurate detection of surfaces and maintains object stability. It continuously tracks device movement and updates object position accordingly.

Functions of this module include:

- Surface detection using ARCore
- Position tracking of objects
- Maintaining stability during movement
- Handling environmental changes

Accurate tracking is essential for delivering a realistic and smooth AR experience.

5.3.4 Reset and Control Module

This module provides control options for managing the application. It allows users to reset the scene, remove objects, and switch between different models.

Key features include:

- Resetting the AR session
- Clearing placed objects
- Changing models
- Reinitializing the environment

This module improves user control and enhances overall usability.

5.4 Testing and Debugging

Testing is an essential part of implementation to ensure that the application functions correctly. The system was tested under different conditions to identify and fix errors.

Testing activities included:

- Checking surface detection accuracy
- Verifying object placement and interaction
- Testing performance on mobile devices
- Identifying and fixing bugs

Debugging was performed using Unity tools and logs to resolve issues related to tracking, performance, and UI.

5.5 Performance Optimization

To ensure smooth operation, performance optimization techniques were applied during development. Since AR applications run in real time, optimization is crucial.

Optimization techniques include:

- Reducing model complexity
- Compressing textures
- Efficient memory management
- Minimizing processing load

These improvements help in reducing lag and ensuring a better user experience.

6. FUTURE SCOPE AND CONCLUSION

6.1 Future Scope

The developed AR application provides a basic framework for interactive visualization, but there are many opportunities for further improvement and expansion. With advancements in technology, the system can be enhanced to support more complex features and real-world applications.

Future improvements can include:

- **Multi-user AR interaction:** Allow multiple users to interact with the same AR environment simultaneously
- **Integration with Artificial Intelligence (AI):** Enable intelligent object recognition and automated responses
- **Improved graphics and realism:** Use advanced rendering techniques for better visual quality
- **Cloud-based data storage:** Store and retrieve AR content dynamically using cloud services
- **Cross-platform compatibility:** Extend support to iOS and AR glasses
- **Voice and gesture control:** Add advanced interaction methods for better usability

In addition, the application can be applied in various fields such as:

- Education for interactive learning
- Healthcare for medical visualization
- Engineering for design and simulation
- Gaming for immersive experiences

These enhancements can make the system more powerful, scalable, and suitable for real-world deployment.

6.2 Conclusion

The internship provided valuable practical exposure to the field of Game Development and Augmented Reality. Through this project, an interactive AR application was successfully developed using Unity, AR Foundation, and ARCore.

The project demonstrated how virtual objects can be integrated into real-world environments and interacted with in real time. It helped in understanding key concepts such as surface detection, object tracking, and user interaction.

The development process improved technical skills in:

- Unity and AR application development
- C# programming
- Debugging and performance optimization
- UI design and user interaction

In addition to technical knowledge, the internship also enhanced professional skills such as communication, teamwork, and project presentation.

Overall, the project achieved its objectives and provided a strong foundation for further learning and development in AR/VR technologies. It also highlighted the importance of combining theoretical knowledge with practical implementation to build effective solutions.

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